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TEMPO: Efficient Test-Time Adaptation For Robot Policies

Critic Entropy Regularization and Score-Based Memory Enable On-Device Policy Adaptation—Up to 80% Performance Recovery on Physical Quadrupeds Under Load

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Reinforcement-learned locomotion policies trained in simulation degrade in deployment when conditions shift—payload changes the center of mass, terrain modifies effective friction, sensors drift with temperature. TEMPO addresses this with a surprisingly effective combination: an entropy regularizer on the critic that prevents value collapse under distribution shift, paired with a score-based memory buffer that upweights hard observations during on-device adaptation. Across three quadruped platforms (Go1, Go2, G1) and four shift types in MuJoCo Playground, TEMPO yields up to 80% improvement in the single-device setting; on a physical Unitree Go2 carrying an added payload, TEMPO recovers pre-shift performance within minutes.

AT A GLANCE

Problem: Sim-to-real locomotion policies degrade when deployment conditions (payload, terrain, sensors) differ from training—and no simulator is available at test time.

Why it matters: Real-world deployment requires policies that adapt autonomously; simulator-based fine-tuning is impractical on hardware.

Method: Critic entropy regularizer prevents value collapse; score-based memory oversamples observations the policy finds hard.

Result: Up to 80% performance improvement in simulation; physical Unitree Go2 recovers from payload shift.

Evidence: Go1, Go2, G1 in MuJoCo Playground under four distribution shifts; physical Unitree Go2 experiment.

The Big Picture

Reinforcement learning in simulation remains the dominant approach for robot locomotion, but the sim-to-real gap means deployed policies face conditions their training never covered. The standard remedy—domain randomisation—widens the training distribution but cannot anticipate all deployment shifts. Once deployed, the robot's own experience is the only data available for further improvement.

Test-time adaptation (TTA) of neural networks has been studied for supervised classification, but RL policies present a harder challenge: there are no ground-truth labels, the reward is expensive to evaluate, the critic can diverge under distribution shift, and compute is constrained to a single embedded device. TEMPO is the first method to address all four constraints simultaneously.

Why Existing Approaches Fall Short

Standard PPO fine-tuning at test time: Without a simulator, PPO must use on-device rollouts directly. The off-policy nature of the limited experience combined with a shifting observation distribution causes the critic to diverge, destabilising policy updates.

Domain randomisation: Broad randomisation reduces peak sim performance and still fails for shifts outside the randomised range (e.g., a specific payload not anticipated during training).

Meta-learning (MAML, PEARL): Meta-learning requires a distribution over adaptation tasks at training time—unavailable when the real-world distribution is unknown—and inner-loop gradient computation that is expensive on embedded hardware.

Core Mechanism

TEMPO adds two components on top of PPO, requiring no changes to the policy network or training procedure:

1. Critic Entropy Regularizer. The critic loss is augmented with an entropy penalty:

$$\mathcal{L}_V = \mathcal{L}_{\text{TD}} + \beta\, H(V_\phi(s))$$

where $H(V_\phi(s))$ is the entropy of the value distribution across a replay batch. This prevents the critic from collapsing to a near-constant value function under distribution shift, which would zero out policy gradient signals.

2. Score-Based Memory. Each observation s in the replay buffer is assigned a score proportional to its TD error:

$$\text{score}(s) \propto |r + \gamma V_\phi(s') - V_\phi(s)|$$

Observations with high scores—those where the critic is most wrong—are sampled more frequently, biasing updates toward the hardest parts of the new distribution. Scores are updated online as the critic improves.

Technical Formulation

The full TEMPO objective at each adaptation step is:

$$\mathcal{L}_{\text{TEMPO}} = \mathcal{L}_{\text{PPO}} + \lambda_V \left[\mathcal{L}_{\text{TD}} + \beta\, H(V_\phi)\right]$$

where $\mathcal{L}_{\text{PPO}}$ is the clipped surrogate objective and λ_V balances the critic and actor losses.

The score-based sampler maintains a priority queue over N stored transitions, replacing the oldest with probability proportional to inverse score (keeping recent hard examples longer). Sampling probability for transition i is:

$$p_i = \frac{\text{score}(s_i)^\alpha}{\sum_j \text{score}(s_j)^\alpha}, \quad \alpha = 0.6$$

(Priority exponent α follows the PER convention.)

Learning or Inference Procedure

Offline training (simulation): Standard PPO with domain randomisation. No modifications to the training procedure.

Online adaptation (deployment): At each deployment step:

1. Execute the current policy on the robot; collect transitions.
2. Update transition scores from critic TD errors.
3. Sample a batch from the priority queue.
4. Apply one PPO + entropy-regularised critic update step.

Total compute per step: equivalent to one standard PPO update, feasible on the onboard CPU/GPU of a Unitree robot.

What the Guarantee Says

Under mild assumptions (bounded reward, Lipschitz value function, score ergodicity), the entropy regularizer prevents the critic from collapsing to a degenerate fixed point during distribution shift:

$$H(V_\phi(s)) \geq H_{\min} > 0 \;\text{ for all } t$$

ensuring policy gradient variance stays non-zero. This provides a necessary condition for adaptation progress; a convergence rate depends on the magnitude of the distribution shift and the data collection speed.

Experimental Findings

- **Go2 (MuJoCo Playground, payload shift):** TEMPO reaches 80% of pre-shift performance within 5 minutes of on-device interaction; standard PPO fine-tuning diverges.
- **Go1, G1 (MuJoCo Playground, four shift types):** Consistent gains across all platforms and shift types; average +42% over non-adapted baseline.
- **Physical Unitree Go2:** TEMPO with a 2 kg added payload recovers stable locomotion within 10 minutes; the non-adapted policy fails to sustain a trot gait.
- **Baseline comparison:** Outperforms domain randomisation, meta-RL (MAML), and test-time normalisation across all conditions.

Ablations and Interpretation

Entropy regularizer only: Without score-based memory, the critic stabilises but adaptation is slower (-25% performance at 5 min); both components are needed.

Score-based memory only: Without entropy regularization, the critic still diverges in 3 of 4 shift scenarios, confirming that stability must come before curriculum selection.

Priority exponent α: Performance is robust to $\alpha \in [0.4, 0.7]$; $\alpha < 0.4$ approaches uniform sampling and loses most gains.

Batch size: Adaptation improves up to batch size 64 and plateaus, consistent with the variance reduction argument underlying critic entropy regularization.

Reference

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